PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Emmanuel John
Program/Batch	B084-Networkwalks
Date	11 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM4 (Footprinting with Harvester) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 4:** Harvester

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purposes only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorized access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.
All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detects whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IPs and MAC addresses.
MacOS Terminal	Local IP and MAC address identification

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the `networkwalks.com` domain using six Kali Linux tools: **WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon**. Each tool was used to collect a different type of information about the target.

First, I used **WHOIS** to obtain publicly available domain registration information and identify the domain's name servers. The results provided information about the domain registration and hosting infrastructure.

I then used **WhatWeb** to identify technologies used by the website. The results identified **WordPress 7.1.4**.

Using **Nslookup**, I resolved the domain name to its IP address. The provided result identified **192.232.216.135**.

I used **Curl** with the `-I` option to inspect the HTTP response headers. This provided additional information about the web application and exposed the WordPress REST API endpoint `/wp-json/`.

Next, I used **Wafw00f** to determine whether a Web Application Firewall was protecting the website. The result identified **ModSecurity (SpiderLabs)**.

Finally, I used **DNSRecon** to enumerate DNS records. The results provided information relating to name servers, mail servers, SPF/TXT records, service records and DNS software information.

4.2 Network Scanning with Zenmap

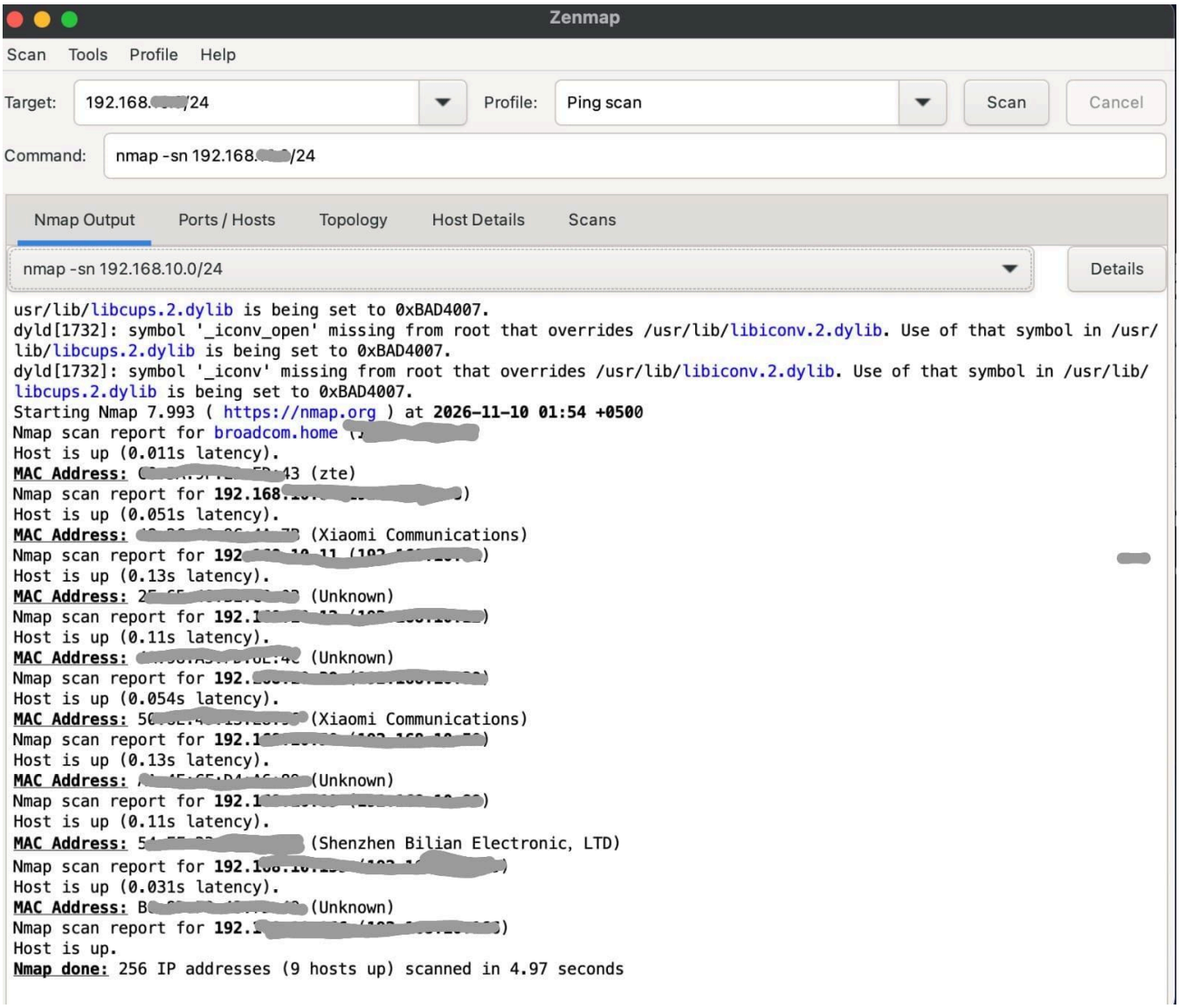
For the second activity, I used **Zenmap** to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the MAC `ifconfig` command to identify my local IP address and LAN subnet. I then entered the subnet into Zenmap and selected **Ping Scan** to identify active hosts.

The example results provided in the practical identified nine live hosts.

The example results also included MAC addresses.

After completing the scan, I opened the **Topology** section in Zenmap, enabled the legend and saved the network topology in PDF format as required by the practical task.



4.3 Footprinting with Harvester

theHarvester is a footprinting tool which is used for gathering information of emails, sub-domains, hosts, employee names, open ports and banners from different public sources like search engines, PGP key servers and SHODAN computer database.

4.3.1 TASK 1

$ theHarvester -d microsoft.com -l 1000 -b baidu
Here -d is the target domain, -l limits the number of results, and -b sets the data source. theHarvester then extracts the details and shows them on screen.

4.3.2 TASK 2

$ theHarvester -d microsoft.com -l 50 -b all
typed above command to start theHarvester & search for the required detail using “all”

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager	Attackers may use exposed technology/version information to identify software requiring further security review	● **Medium**
2	Server IP address identifiable	Nslookup resolved the domain to `192.232.216.135`	Provides information about the network location of the web service	● **Low**
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed `/wp-json/`	May assist technology fingerprinting and further enumeration	● **Low**
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application’s security architecture	● **Low**
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	● **Medium**
6	Multiple live hosts visible on local network	Zenmap identified nine live hosts in the example network	Unknown or unauthorized devices may potentially be present on a network	● **Medium**

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information**
   Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.
2. **Keep software updated**
   CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers**
   HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Review DNS records regularly**
   DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF**
   Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Perform regular internal network discovery**
   Organizations should periodically scan their own networks to identify active devices.
7. **Investigate unknown devices**
   Any unexpected device discovered during network scanning should be investigated and verified.
8. **Maintain network documentation**
   Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization**
   Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

In the network scanning activity, I used Zenmap to identify my local network configuration and discover active hosts. I also collected IP and MAC address information and created a network topology.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

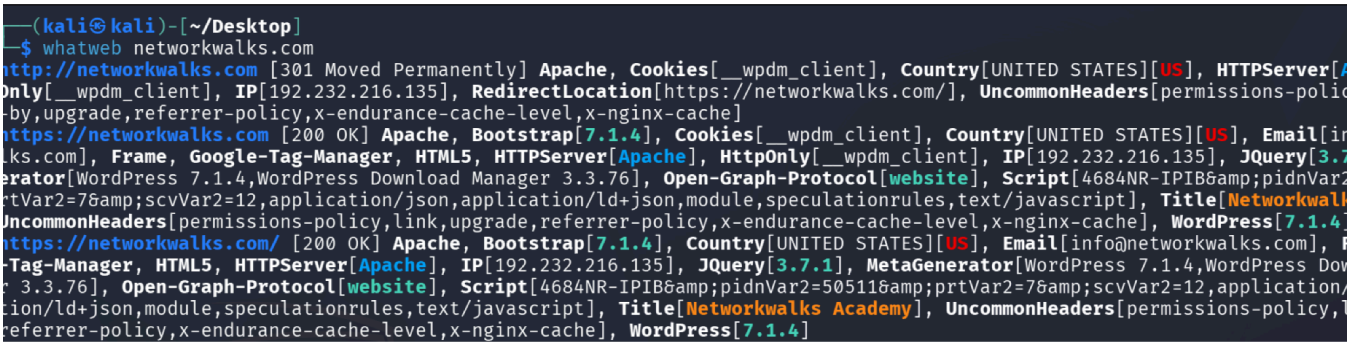
8. Evidence

```
┌──(kali㉿kali)-[~/Desktop]
└─$ whois networkwalks.com
   Domain Name: NETWORKWALKS.COM
   Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.godaddy.com
   Registrar URL: http://www.godaddy.com
   Updated Date: 2025-11-12T10:08:43Z
   Creation Date: 2019-11-06T22:51:46Z
   Registry Expiry Date: 2027-11-06T22:51:46Z
   Registrar: GoDaddy.com, LLC
   Registrar IANA ID: 146
   Registrar Abuse Contact Email: abuse@godaddy.com
   Registrar Abuse Contact Phone: 480-624-2505
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
   Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
   Name Server: NS6135.HOSTGATOR.COM
   Name Server: NS6136.HOSTGATOR.COM
   DNSSEC: unsigned
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-11-09T21:52:00Z <<<

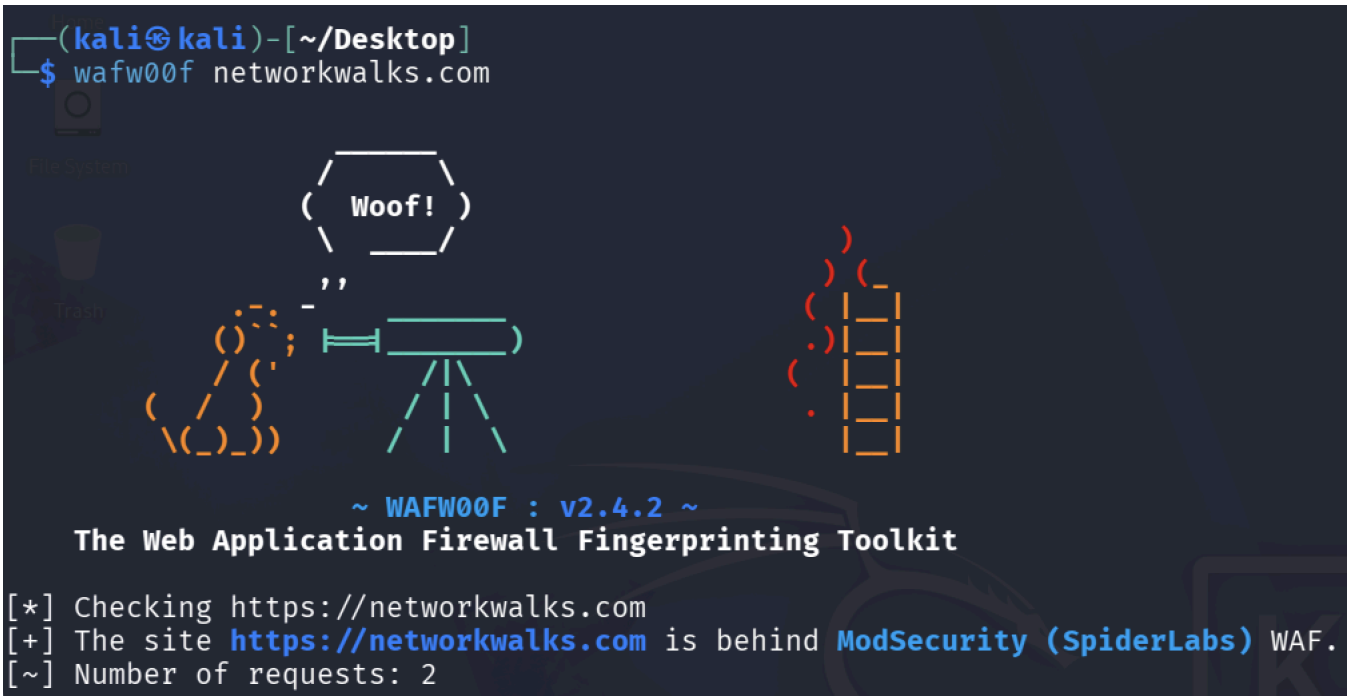
For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar.  Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.
```

Attacker Perspective: WHOIS reveals domain registration information. The name servers point to HostGator, revealing the hosting provider. Registration and expiry dates help with social engineering and domain expiration attacks.



Attacker Perspective: WhatWeb exposes the exact software and versions. Attackers check these versions in vulnerability databases to find known exploits. It also reveals the server IP address.



Attacker Perspective: Wafw00f tells an attacker if a firewall is watching. Knowing a WAF is present means naive attacks will be blocked or logged, so attackers must adapt or try to bypass it.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ nslookup networkwalks.com
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```

Attacker Perspective: Nslookup resolves the domain to its real IP address. Knowing the IP lets an attacker scan the server directly, find other sites on the same IP, and map the target's infrastructure.

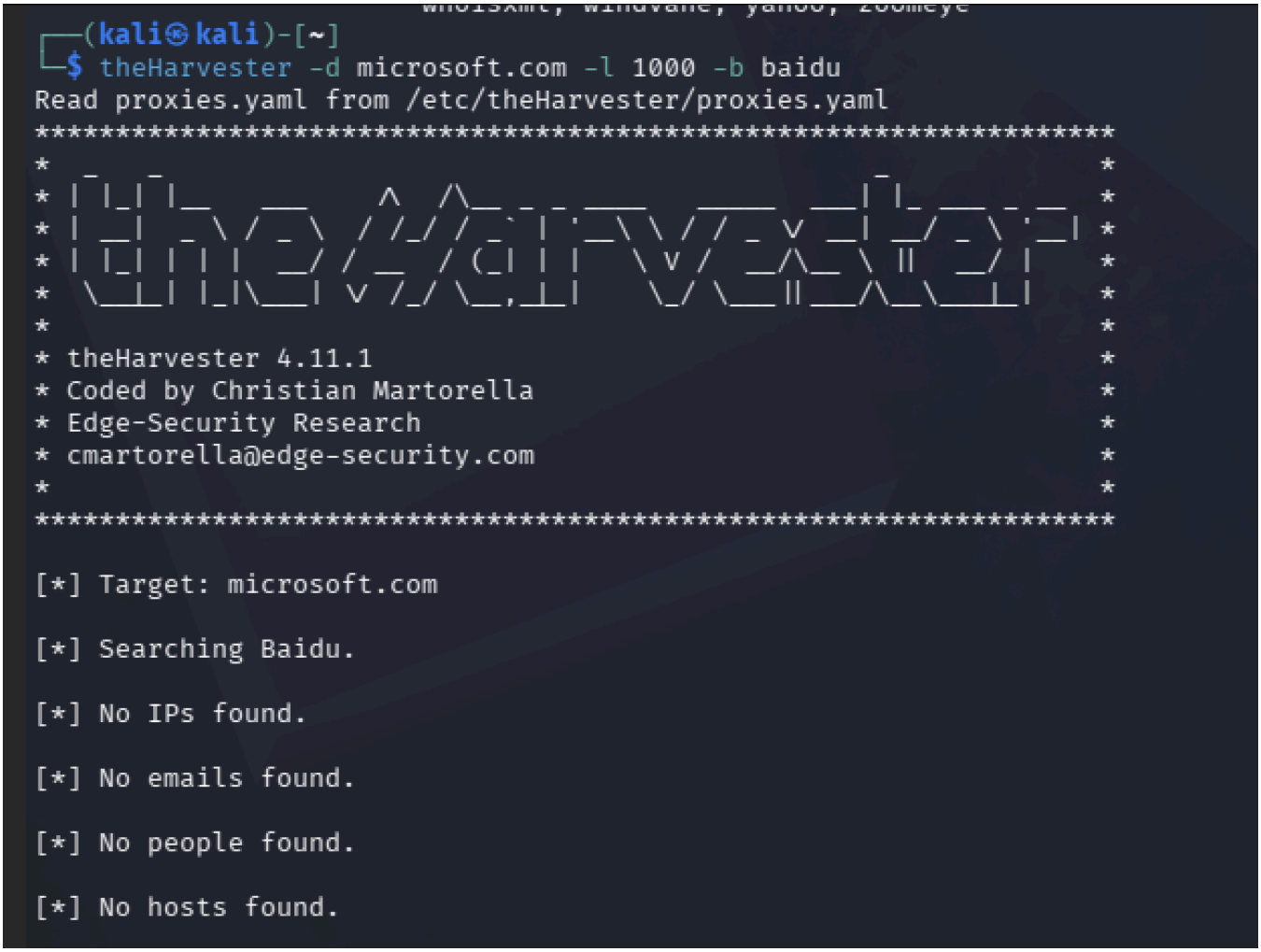
```
┌──(kali㉿kali)-[~/Desktop]
└─$ dnsrecon -d networkwalks.com
2026-11-09T17:20:06.577819-0500 INFO Starting enumeration for domain: networkwalks.com
2026-11-09T17:20:06.578138-0500 INFO std: Performing General Enumeration against: networkwalks.com...
2026-11-09T17:20:07.394881-0500 ERROR No answer for DNSSEC query for networkwalks.com
2026-11-09T17:20:08.124077-0500 INFO     SOA ns6135.hostgator.com 50.87.144.87
2026-11-09T17:20:09.134970-0500 INFO     NS ns6136.hostgator.com 192.232.216.131
2026-11-09T17:20:09.728417-0500 INFO     Bind Version for 192.232.216.131 "9.16.23-RH"
2026-11-09T17:20:09.728698-0500 INFO     NS ns6135.hostgator.com 50.87.144.87
2026-11-09T17:20:10.374003-0500 INFO     Bind Version for 50.87.144.87 "9.16.23-RH"
2026-11-09T17:20:11.501674-0500 INFO     MX mail.networkwalks.com 192.232.216.135
2026-11-09T17:20:11.624769-0500 INFO     A networkwalks.com 192.232.216.135
2026-11-09T17:20:12.627539-0500 INFO     TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-11-09T17:20:12.629440-0500 INFO     TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-11-09T17:20:13.264547-0500 INFO Enumerating SRV Records
2026-11-09T17:20:16.408291-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.14 443
2026-11-09T17:20:16.408472-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.11 443
2026-11-09T17:20:16.408561-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.9 443
2026-11-09T17:20:16.408659-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.15 443
2026-11-09T17:20:16.408747-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.11 443
2026-11-09T17:20:16.408811-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.9 443
2026-11-09T17:20:16.408868-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.14 443
2026-11-09T17:20:16.408927-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.15 443
2026-11-09T17:20:16.410444-0500 INFO 8 Records Found
2026-11-09T17:20:16.410681-0500 INFO Completed enumeration for domain: networkwalks.com
```

Attacker Perspective: DNSRecon maps the entire DNS footprint. Each record is a potential foothold and helps attackers understand the email and hosting setup.

```
┌──(kali㉿kali)-[~/Desktop]
└─$  curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net"
"https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://ww
w.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alterna
te"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=e1ead4336f9117bbc50484fbd8ef8822; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Mon, 09 Nov 2026 22:15:05 GMT
server: Apache
```

Attacker Perspective: HTTP headers leak the web server, caching stack and hidden endpoints like the WordPress REST API. Attackers read headers to fingerprint the stack and find entry points without even loading the full page.

Harvester Result



```
┌──(kali㉿kali)-[~/Desktop]
└─$ theHarvester -d microsoft.com -l 50 -b all
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*******************************************************************
*                                                                 *
*  _   _                                            _             *
* | |_| |__   ___    /\  /\__ _ _ ____   _____  ___| |_ ___ _ __  *
* | __| '_ \ / _ \  / /_/ / _` | '__\ \ / / _ \/ __| __/ _ \ '__| *
* | |_| | | |  __/ / __  / (_| | |   \ V /  __/\__ \ ||  __/ |    *
*  \__|_| |_|\___| \/ /_/ \__,_|_|    \_/ \___||___/\__\___|_|    *
*                                                                 *
* theHarvester 4.11.1                                             *
* Coded by Christian Martorella                                   *
* Edge-Security Research                                          *
* cmartorella@edge-security.com                                   *
*                                                                 *
*******************************************************************

[*] Target: microsoft.com

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverun search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bufferoverun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform BuiltWith search for word: 'microsoft.com'
A Missing Key Error occurred in builtwith:
[!] Missing API key for BuiltWith.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process brave search for word: 'microsoft.com'
Error Message:
[!] Missing API key for Brave Search.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Censys API key is missing or invalid:
[!] Missing API key for Censys ID and/or Secret.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing key error occurred in criminalip:
[!] Missing API key for criminalip.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in dehashed:
[!] Missing API key for Dehashed.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for DNSDumpster.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in dymo:
[!] Missing API key for dymo.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
A Missing Key error occurred in fullhunt:
[!] Missing API key for fullhunt.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Error initializing SearchGithubCode:
[!] Missing API key for Github.
A Missing Key error occurred in github-code:
[!] Missing API key for Github.
```

```
[*] No LinkedIn users found.

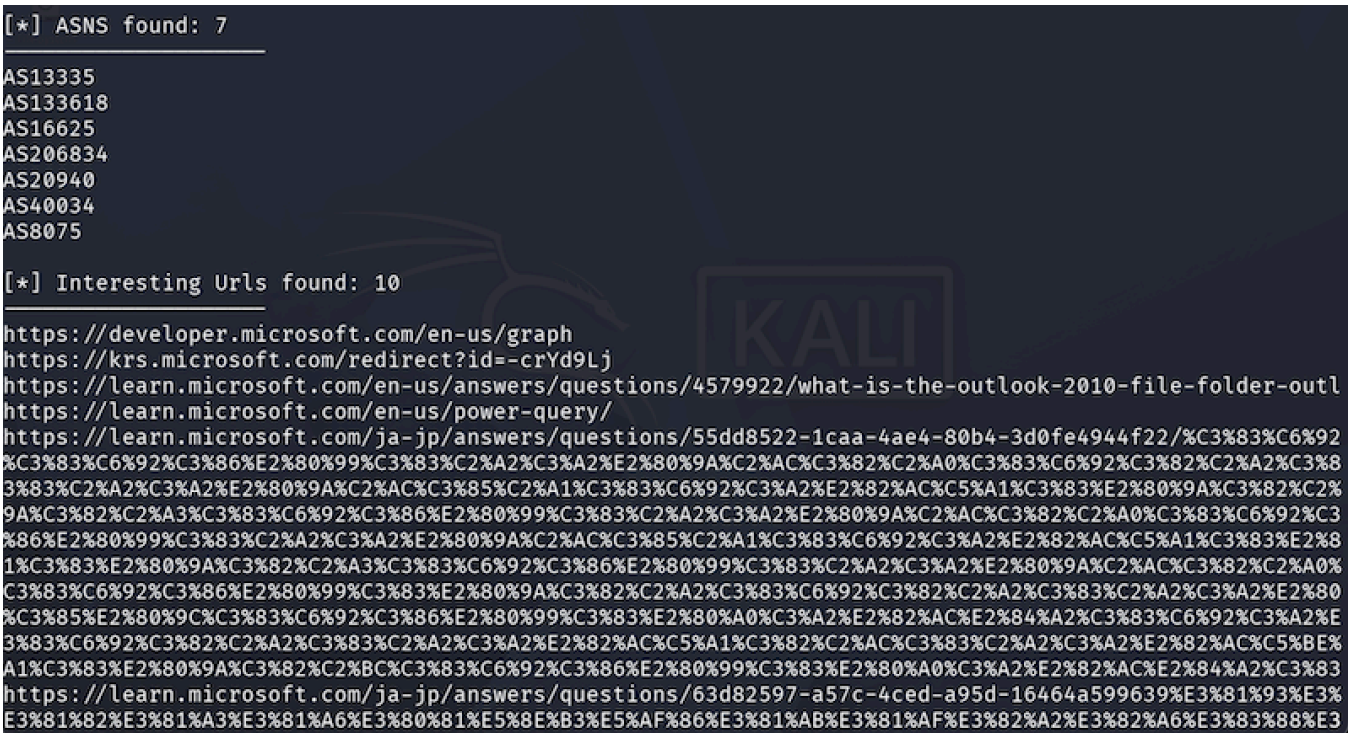

[*] LinkedIn Links found: 0
──────────────────────────

[*] IPs found: 151
──────────────────────────
103.224.182.247
103.224.182.252
103.224.212.141
103.224.212.142
104.18.41.41
104.215.148.63
104.40.211.35
104.43.195.251
13.107.137.11
13.107.139.11
13.107.213.40
13.107.213.69
13.107.226.35
13.107.226.40
13.107.226.41
13.107.226.67
13.107.226.69
13.107.226.70
13.107.226.71
13.107.246.40
13.107.246.69
13.107.253.35
13.107.253.40
13.107.253.41
13.107.253.67
13.107.253.69
13.107.253.70
13.107.253.71
13.77.161.179
134.170.185.46
134.170.188.221
134.170.188.232
150.171.109.100
150.171.109.101
150.171.109.106
150.171.109.113
150.171.109.114
150.171.109.115
150.171.109.116
150.171.109.117
150.171.109.118
150.171.109.119
150.171.109.132
150.171.109.145
150.171.109.146
150.171.109.147
150.171.109.148
150.171.109.149
```

```
[*] Emails found: 3
-----------------------
dotnet-docker-bot@microsoft.com
opencode@microsoft.com
secure@microsoft.com

[*] No people found.

[*] Hosts found: 9490
---------------------
*.api.cfp.microsoft.com
*.bigcatalog-ppe.commerce.microsoft.com
*.bizextsvc-int.microsoft.com
*.canary.mfavalidation.auth.microsoft.com
*.canary.testmymfa.auth.microsoft.com
*.ckms.prod.communication.microsoft.com
*.cloudcare-aks-ppe.caresupplychain.microsoft.com
*.cloudcare-aks-prd.caresupplychain.microsoft.com
*.console-m365d.microsoft.com
*.copilotstudio.microsoft.com
*.dl.adu.microsoft.com
*.emailinbg.ppe.communication.microsoft.com
*.events.prod.communication.microsoft.com
*.frontdoor.bigcatalog-ppe.commerce.microsoft.com
*.preprod.api.bap.microsoft.com
*.prod.api.copilotforfinance.microsoft.com
*.publisher-aircapi.1pp.microsoft.com
*.rooms.ppe.communication.microsoft.com
*.test.api.bap.microsoft.com
*.tip1.api.bap.microsoft.com
*.tip2.api.bap.microsoft.com
04bb462a-3730-48e5-84db-56f42351ec71.microsoft.com
05c50d84-b811-498b-b2d0-d720f443091c.microsoft.com
064-smtp-in-2a.microsoft.com
064-smtp-in-2a.microsoft.com:157.54.41.37
0688854f-d5d9-4885-8447-8599aa383f4e.microsoft.com
08dbb1e2-14d0-4668-a751-5e89b2ecdc61.microsoft.com
0c1bc83d-d1fe-445b-8346-09c567e07385.microsoft.com
11b59461-755a-4882-8ba9-58e2e788d644.microsoft.com
140.82.8.34.microsoft.com
149.28.32.9.microsoft.com
14cd4cb2-348e-4864-8cd9-de45179cc5d2.microsoft.com
17e3e2bf-2dd9-4ed3-8b46-bed1c6112784.microsoft.com
1df037eb-e2f8-4337-9ecb-1e61a37c1c20.microsoft.com
1eec5887-103c-48d0-a492-c2a9945d4845.microsoft.com
1es-mcp-gallery.microsoft.com
1es-mcp-gallery.microsoft.com:150.171.110.17
1esbot-test.microsoft.com
1esbot.microsoft.com
1eslivesecrets.microsoft.com
1esrm.tme.cloudtest.microsoft.com
```



Attacker Perspective: TheHarvester collects information from public sources without touching the target directly. Each harvested email is a potential target for phishing. Each subdomain is another door into the organization. Because it only reads public data, the target never knows it is being studied.

-End-

Author
Saba Fatima
Cybersecurity Professional B084
LinkedIn: https://www.linkedin.com/in/sabafatimamin/

Project Information
Program Name: Cybersecurity program at Networkwalks | **Week:** 02 |
Repository: Github